BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 4 | Duration: 40 min
Semester: Spring 2025 | Full Marks: 15

CSE 470: Software Engineering

Name: **ID:** **Section:**

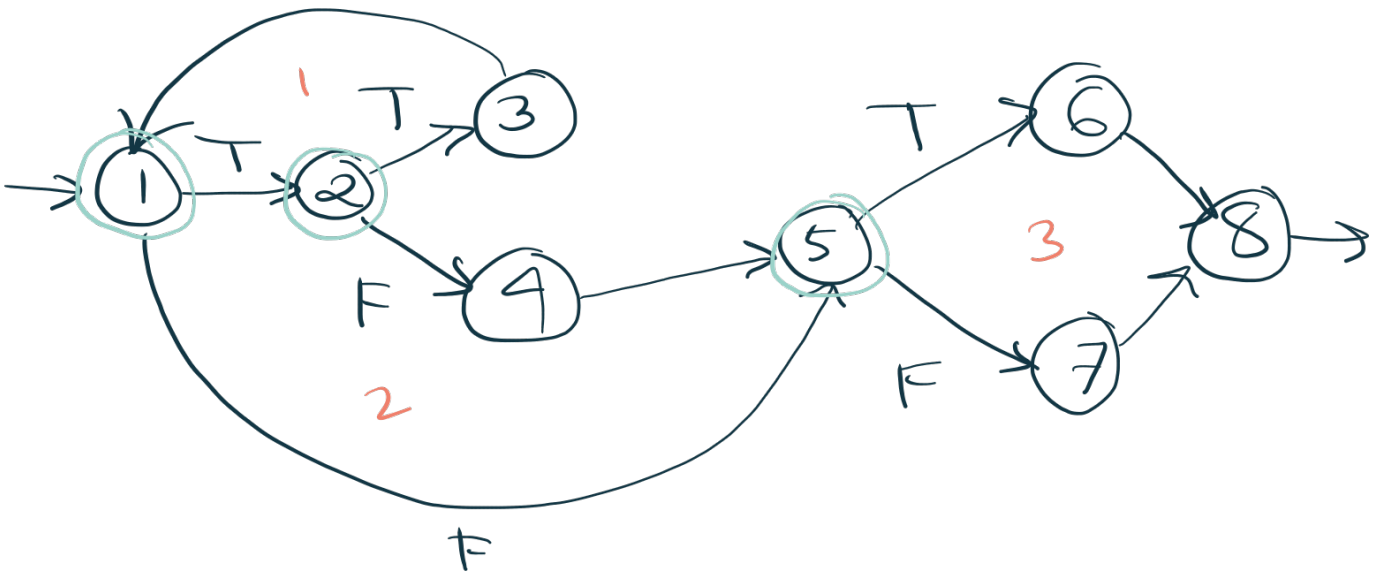
Python	Java

def move_to_station(distance, battery):
  while distance > 0:
  if battery > 10:
  print(f"Moving... distance left: {distance}")
  distance -= 1
  battery -= 5
  else:
  print("Low battery! Stopping.")
  break
 	
  if distance == 0:
  print("Reached charging station.")
  else:
  print("Could not reach charging station.")
	public class Robot {
	  public static void moveToStation(int distance, int battery) {
	  while (distance > 0) {
	  if (battery > 10) {
	  System.out.println("Moving... distance left: " + distance);
	  distance--;
	  battery -= 5;
	  } else {
	  System.out.println("Low battery! Stopping.");
	  break;
	  }
	  }
	
	  if (distance == 0) {
	  System.out.println("Reached charging station.");
	  } else {
	  System.out.println("Could not reach charging station.");
	  }
	  }
	}

[You must mark which code block/line corresponds to which node in the question paper.]

1.	**CO3**	a. Draw the CFG for the given method [5 Marks] b. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set [3 + 3 = 6 Marks] c. Generate test cases for all paths in the Basic Paths Set [4 Marks]	**15**

a)



b) M = R+1 = 3+1 = 4

P+1 = 3+1 = 4

E - N + 2P = 10 - 8 + 2(1) = 4

Paths:

① 1 - 2 - 3 - 1 - 5 - 6 - 8

② 1 - 2 - 4 - 5 - 7 - 8

Test Cases

① distance = 1 battery = 20

Output: Moving... distance left: 0

Reached charging station

② distance = 2 battery = 5

Output Low battery! Stopping.

Could not reach charging station